

Class 2 SP250: Primes & Composites with Euclid's proof of infinite primes

More Divisibility - Primes and Composites

b is called a **divisor** of a if there exists an integer c such that $a = bc$.
Because $n = 1 * n$, any integer n is divisible by itself and by 1.

Definition: A positive integer n is called a **prime number** if it has exactly two divisors: itself and 1.
A positive integer n is called **composite** if it has more than two positive integer divisors.
1 is neither prime nor composite.
Any integer $n > 1$ has at least one prime divisor.

The first few prime numbers are: 2, 3, 5, 7, 11, 13, 17,

Question: How many prime numbers are even?

Answer: there is exactly one, namely 2. All other even numbers are divisible by 2 by definition of even and thus have at least 3 divisors.

What is the highest prime number that has been discovered? $2^{82,589,933} - 1$ - it has 24,862,048 digits!

How many prime numbers are there?

Theorem (Euclid ca. 300BC): There are infinitely many primes.

Argue by contradiction.

- Assume that there are finitely many primes and that they are $p_1=2, p_2=3, \dots, p_k$. All primes are in this set.

Super, super clever idea:

Consider the number:

$$N = p_1 * p_2 * p_3 * \dots * p_k + 1 !!!$$

By our assumption, this number is composite. What are prime divisors of **N**?

- Is N divisible by p_1 ? Why or why not?
N is not divisible by p_1 , because it is one more than a multiple of p_1 .
- So, is N divisible by any of the $p_1, p_2, \dots, p_k$?
No.

Where is the contradiction?

Well, what are prime divisors of N?

There aren't any because $p_1, p_2, \dots, p_k$ are all the primes and N isn't divisible by any of them.

So... N has no prime divisors! Contradiction.

Or... is it?

Why must every number larger than 1 have a prime divisor?

- N is either prime or composite.
If N is prime, then we are done. We have found a prime that is not on the list.
If N is composite, it has divisors other than 1 and N.
- Why does N have a **prime** divisor?
Take the smallest divisor of N, that is larger than 1. Call this q.
q can not be composite because then you could factor it and get a smaller divisor. Hence it's prime. We're done.

Long rows of consecutive numbers

Are there 1,000 numbers in a row, none of which are prime?

Consider $1 * 2 * 3 * \dots * 1000 * 1001 = 1001!$

$1001! + 2, 1001! + 3, 1001! + 4, \dots, 1001! + 1000, 1001! + 1001.$

None of these are prime!!!!